

A FAIRNESS QUESTION FOR TWO INVOLUTIONS ON LATIN SQUARES

SETTING

Let L be a Latin square of order n on the symbol set $[n]$, with rows and columns indexed by $[n]$. Two rows, 1 and 2, and two columns, $j \neq j'$, are distinguished throughout.

The frame. Let $\pi = \pi_L$ be the permutation of the rows defined by

$$L(\pi(r), j') = L(r, j) \quad (r \in [n]),$$

i.e. $\pi(r)$ is the row in which column j' holds the symbol that column j holds in row r . (π is a derangement.) The *bit* of L is

$$\beta(L) = \mathbf{1}\{\text{rows 1 and 2 lie in a common cycle of } \pi_L\}.$$

Row-pair permutations. For rows $x \neq y$ let $\rho_{x,y}$ be the permutation of the columns defined by $L(y, \rho_{x,y}(q)) = L(x, q)$. For a cycle Q of $\rho_{x,y}$ the two rows carry the same set of symbols on the columns of Q .

Two moves.

- (TRADE) For rows $x \neq y$ with $x, y \notin \{1, 2\}$, let Q be the cycle of $\rho_{x,y}$ containing j' , and exchange the entries of rows x and y in the columns of Q .
- (TURN) For a cycle c of π_L containing neither row 1 nor row 2, exchange the entries of columns j and j' in the rows of c .

Lemma 1. *Both moves produce Latin squares, are involutions, and leave rows 1 and 2 unchanged. A turn inverts π on the cycle c and leaves its cycle partition unchanged. A trade at (x, y) replaces π by $(xy) \circ \pi$ if $j \notin Q$ (so the cycles of x and y are merged if distinct and split if equal), and by $(xy) \circ \pi \circ (xy)$ if $j \in Q$ (the cycle partition is unchanged up to exchanging the labels x, y). In particular a trade can change β only when $j \notin Q$; call such a pair (x, y) flippable. Writing $b' = L(y, j')$ and q^* for the column in which row x holds b' , the pair (x, y) is flippable iff q^* does not lie on the $\rho_{x,y}$ -cycle of j ; if $\pi(x) = y$ or $\pi(y) = x$ it is never flippable.*

The proofs are direct verifications. (Both rows carry the same symbols on Q , and both columns carry the same symbols on c , which gives the Latin property; the frame formulas follow from the definition of π , since only the entries of rows x, y in columns j, j' matter.)

COMPONENTS AND THE QUESTION

Fix a $2 \times n$ Latin rectangle R (the first two rows) and columns $j \neq j'$ such that

$$(1) \quad R(1, j) \neq R(2, j') \quad \text{and} \quad R(1, j') \neq R(2, j),$$

i.e. $\pi(1) \neq 2$ and $\pi(2) \neq 1$ for every extension of R . Let $S(R)$ be the set of Latin squares whose first two rows are R . The two moves act on $S(R)$; let $\mathcal{K}(R)$ be the set of orbits

(“components”) of the group they generate, and for a component K let

$$p(K) = \frac{|\{L \in K : \beta(L) = 1\}|}{|K|}.$$

Question 1 (component fairness). *Is it true that, uniformly over R and over column pairs (j, j') satisfying (1),*

$$\frac{1}{|S(R)|} \sum_{K \in \mathcal{K}(R)} |K| (2p(K) - 1)^2 \longrightarrow 0 \quad (n \rightarrow \infty)?$$

Equivalently: for L uniform in $S(R)$, the conditional probability that $\beta(L) = 1$ given the component of L is $\frac{1}{2} + o(1)$ in L^2 . A quantitative version asks for the left side to be $o(1/\log^2 n)$.

Remark 1 (why the question is natural). Under the uniform measure on $S(R)$ the moves define a symmetric Markov chain (pick a uniform pair $x, y \notin \{1, 2\}$ and trade, or a uniform row $x \notin \{1, 2\}$ and turn its cycle when allowed), so the uniform measure on each component is stationary and the left side of the Question is $\lim_{M \rightarrow \infty} \mathbb{E}[\beta'(L_0)\beta'(L_{2M})]$ with $\beta' = 2\beta - 1$, the long-time autocorrelation of the bit. Along the chain the frame π performs the walk of left multiplication by transpositions (xy) , $x, y \notin \{1, 2\}$, *thinned* to the flippable pairs (the turns and the non-flippable trades only relabel rows within cycles). The unthinned walk preserves $\pi^{-1}(1)$ and $\pi^{-1}(2)$ and is the random-transposition walk on S_{n-2} after contracting the arcs $\pi^{-1}(1) \rightarrow 1$ and $\pi^{-1}(2) \rightarrow 2$; on each of its cosets exactly half of the permutations have rows 1, 2 in a common cycle. So if every pair were flippable the answer would be yes, with rate $o(1)$ after $O(n \log n)$ steps by Diaconis–Shahshahani. The content of the Question is that the thinning by flippability does not destroy this.

Remark 2 (what is known numerically). On uniformly random Latin squares with $n = 30$ –100, a uniformly random pair $x, y \notin \{1, 2\}$ is flippable with probability 0.50 ± 0.01 , independently of the relative position of x, y in the cycles of π except for the exclusion $\pi(x) = y$, $\pi(y) = x$; the fraction of flippable pairs in a square is concentrated (0.42–0.55 at $n = 30$); and the autocorrelation of the bit along the chain, as a function of the number of trades performed, follows that of the unthinned walk and decays to zero. No square with $|2p(K) - 1|$ bounded away from 0 has been observed except for components of trivial size.

Remark 3 (weaker forms that are still useful). (i) A yes in expectation over R — for (R, j, j') drawn as the first two rows and a column pair of a uniformly random Latin square, with (j, j') restricted to pairs at a prescribed distance along a cycle of the row permutation $R(1, \cdot) \mapsto R(2, \cdot)$ — suffices for the application below. (ii) A sufficient condition is that, for all but an $o(1/\log^2 n)$ -fraction of $L \in S(R)$, a positive proportion of the bit-changing pairs (the pairs x, y in the common cycle of rows 1, 2 that separate them when $\beta = 1$; the pairs $x \in C_1$, $y \in C_2$ when $\beta = 0$) are flippable, together with a mixing estimate for the thinned walk.

Remark 4 (provenance). The Question arose in a programme on the Cavenagh–Greenhill–Wanless conjecture on the cycle type of the second row of a random Latin square: with (j, j') a marked column pair $\{j, \sigma^\alpha(j)\}$, $\alpha \geq 2$, inside a cycle of the row permutation σ , the marked-pair identity $2\mathcal{C}_n(\lambda)/\mathcal{C}_n(\mu) - 1 = \mathbb{P}[\beta = 1]/\mathbb{P}[\beta = 0]$ reduces the conjecture to $\mathbb{P}[\beta = 1] = \frac{1}{2} + o(1/\log n)$ over such instances, and a positive answer to the quantitative Question, averaged as in Remark 3(i), gives exactly that. The question itself makes no reference to this.